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2023–24 Private School Universe Survey (PSS) Public-Use Data File User's Manual

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Ⅰ. Overview

The Private School Universe Survey (PSS) is conducted by the National Center for Education Statistics (NCES) on behalf of the U.S. Department of Education to collect basic information on American private elementary and secondary schools. PSS grew out of a proposal, in 1988, to develop a private school data collection that would improve the sporadic collection of private school data dating back to 1890 and improve commercially available private school sampling frames. PSS was first collected by the U.S. Census Bureau in the 1989–90 school year, with data collections every 2 years since.

Target Population

The target population for PSS is all schools in the United States that are not supported primarily by public funds, provide classroom instruction for one or more of grades kindergarten through 12 (or comparable ungraded levels), and have one or more teachers. Organizations or institutions that provide support for homeschooling, but do not provide classroom instruction, are not included.

Purpose and Content of the Survey

PSS is currently designed to generate biennial data on the total number of private schools, teachers, and students, and to build a universe of private schools to serve as a sampling frame for NCES sample surveys.

The 2023–24 PSS instrument consisted of a single school questionnaire designed to be filled out by school administrators. Data collected included enrollment by grade, enrollment by race/ethnicity and sex, number of high school graduates, number of teachers, program emphasis, school religious orientation or affiliation, association membership, existence and type of kindergarten program, number of days in the school year and length of the school day, and whether the school had a library media center. For the 2023–24 PSS questionnaire, a question was added about virtual and hybrid schooling.

Content of the User's Manual

This manual contains chapters covering frame creation, data collection and response rates, data preparation and imputation, guide to the public-use data file and codebook, and user notes and cautions.

II. Frame Creation

Since PSS attempts to include every private school in the United States, a universe list of private schools meeting the PSS definition (i.e., not supported primarily by public funds, providing classroom instruction for one or more of grades kindergarten through 12 or comparable ungraded levels, and having one or more teachers) must be created. Since 1983, NCES has used a dual frame approach for building the private school universe (Beller 1984). The dual frame consists of a list frame and an area frame. The combination of the list frame schools, and additional schools identified in the area search, comprised the schools included in the 2023–24 PSS. Because PSS uses a dual frame approach, it is possible to estimate the coverage or completeness of PSS. Using the most conservative approach for estimation, the coverage rate of traditional private schools is estimated to be 73.14 percent.

List Frame

The list-building component was the primary means for improving coverage of private schools. The basis for the 2023–24 PSS list frame was the 51,402 private schools from the 2021–22 PSS list and area frames. To provide coverage of private schools founded since 2021 and to improve coverage of private schools existing in 2023, the Census Bureau requested membership lists from 30 private school associations and religious denominations; 21 of these lists were received and processed. The departments of education from each of the 50 states and the District of Columbia were asked to provide lists of private schools. There were 48 states lists received and processed (lists were not received from Arizona, the District of Columbia, and Oklahoma). Additionally, each state and the District of Columbia were asked to provide lists of early childhood centers to identify k-terminal schools (schools that offer kindergarten and no other grade between 1 and 12), and 22 lists were received and processed. Private schools that submitted requests directly to NCES to be included in PSS were added to the list frame.

Schools on private school association membership lists, the state lists, and the early childhood lists were compared to the base list, and any school that did not match a school on the base list was added to the private school universe list. The base list is comprised of private schools that were included on the frame for the prior data collection cycle, PSS 2021–22. Note that the base list includes both active and inactive cases. Inactive cases include schools that were identified as duplicates, no longer in operation, or otherwise out-of-scope and thus ineligible prior to data collection. The total number of private schools added by the 2023–24 list frame operation was 5,273 resulting in a total list frame of 56,675 schools.

Area Frame

To identify private schools that might have been overlooked in the list frame operation, a group of geographic areas was selected to be searched for private schools. The United States is divided by the Census Bureau into 2,062 primary sampling units (PSUs), each PSU consisting of a single county, independent city, or group of contiguous counties. The eight PSUs (Los Angeles and Orange Counties, California; Harris County, Texas; Cook County, Illinois; Queens and Kings Counties, New York; Miami-Dade County, Florida; and Philadelphia County, Pennsylvania) with the highest private school enrollment were selected for the sample with certainty, and have

been in sample since the 2009–10 PSS cycle.[1] An additional 116 non-certainty PSUs were selected by stratified random sampling from the 2,054 remaining non-certainty PSUs using American Community Survey (ACS) five-year data. Due to a 3-year lag, 2016–20 ACS data were used for the 2023–24 PSS.

A total of 124 distinct PSUs (220 counties) were selected for the 2023–24 PSS to create the Area Search Frame (ASF). Within each of these PSUs, the Census Bureau attempted to find all eligible private schools. A block-by-block listing of all private schools in a sample of PSUs was not attempted. Rather, Census Bureau staff created the area frame by using sources such as online yellow pages (yp.com), local Catholic dioceses, local education agencies, and local government offices. The Census Bureau also obtained a list of eligible private schools for area frame sample PSUs from the Data Axle database[2] and a supplemental vendor purchase. Once the area search lists were constructed, they were matched with the NCES private school universe list. Schools already on the universe list were deleted from the area frame. A total of 590 schools were added to the universe from the area frame in 2023–24.

Weighting

The final weights are needed to have the estimates reflect the population of private schools when analyzing the data. The data from the area frame component were weighted to reflect the sampling rates (probability of selection) of the PSUs. Survey data from both the list and area frame components were adjusted for school nonresponse. The final weight for PSS data items is the product of the **Base Weight** and the **Nonresponse Adjustment Factor**, where:

> **Base Weight** is the inverse of the probability of selection of the school. The base weight is equal to one for all list-frame schools. For area-frame schools, the base weight is equal to the inverse of the probability of selecting the PSU in which the school resides.
>
> **Nonresponse Adjustment Factor** is an adjustment that accounts for school nonresponse. It is the weighted (base weight) ratio of the total eligible in-scope schools (interviewed schools plus noninterviewed schools) to the total responding in-scope schools (interviewed schools) within cells. Noninterviewed and out-of-scope cases are assigned a nonresponse adjustment factor of zero.

The cells used to compute the nonresponse adjustment were defined differently for list-frame and area-frame schools. For schools in the list frame, the cells were defined by affiliation (17 categories), locale type (4 categories), grade level (4 categories), Census region (4 categories), and enrollment (up to 20 categories). The nonresponse adjustment cells for area-frame schools were defined by three-level typology (3 categories) and grade level (4 categories).

If the number of schools in a cell was fewer than 15 or the nonresponse adjustment factor was greater than 1.5, then that cell was collapsed into a similar cell. The variables used to collapse the cells and the collapse order varied according to whether the school was from the list or area frame and whether a school was a traditional or k-terminal school. The cells for traditional schools from the list frame were collapsed within enrollment category, locale type, grade level,

[1] The PSS sample PSUs were selected systematically with probabilities proportional to the square root of the 2010 American Community Survey PSU private school enrollment.
[2] See https://www.dataaxleusa.com/.

and Census region. Cells for k-terminal schools from the list frame were collapsed within enrollment category, locale type, Census region, and affiliation. Cells for traditional schools from the area frame were collapsed within grade level and then within three-level typology. Cells for k-terminal schools from the area frame were collapsed within three-level typology.

Variance Estimation

In surveys with complex sample designs, direct estimates of sampling errors that assume a simple random sample will typically underestimate the variability in the estimates. The PSS sample design and estimation included procedures that deviate from the assumption of simple random sampling, primarily resulting from the stratified cluster sampling occurring in the area frame.

Producing Replicate Weights

The preferred method of calculating sampling errors to reflect these aspects of the complex sample design of PSS is using replication. Replication methods involve constructing subsamples, or replicates, from the full sample and computing the statistic of interest for each replicate. The mean square error of the replicate estimates around the full sample estimate provides an estimate of the variance of the statistic.

The replicate weights are used to compute the variance of a statistic, Y, as given in equation 1, where Y_r equals the estimate of Y using the rth set of replicate weights and n equals the number of replicates.

$$\text{Equation 1: Variance } (Y) = \left(\frac{1}{n}\right)\sum_r (Y_r - Y)^2$$

PSS uses a procedure known as balanced repeated replication (BRR) for the calculation of sampling variance. BRR assumes sampling is done with replacement, and hence, BRR does not reflect the increase in precision due to sampling a large proportion of a finite population.

To execute the BRR procedure, half-samples are defined by pairing sample PSUs within each sampling stratum, forming variance strata. The final product is 88 replicate weights. After the variance strata were assigned, an orthogonal matrix (matrix H where: $HH^T = nI_n$ where I_n is the identity matrix of order n) was used to form the 88 balanced half-sample replicates.

Applying Replicate Weights

Each PSS data file includes a set of 88 replicate weights designed to produce variance estimates. Replicate weights were created for each of the 88 samples using the same estimation procedures used for the full sample and are included in the data file. The replicate weights for PSS are REPW1 to REPW88.

Computation of sampling errors using these replicate weights can be done easily using one of the following software: SAS, Stata, SUDAAN, R, AM Statistical Software,[3] or WesVar Complex Sample Software.

[3] Users of AM Statistical Software or WesVar Complex Sample Software can see the PSS 2021–22 user manual for more information.

SAS. Standard errors based on BRR can be generated in SAS version 9.2 or higher using the survey procedures (SURVEYMEANS, SURVEYFREQ, SURVEYREG, and SURVEYLOGISTIC) available in the SAS/STAT module. (This module is an addition to the Base SAS software). The BRR method is specified in the PROC statement using the VARMETHOD option parameter (VARMETHOD=BRR). The replicate weight variables are specified using the REPWEIGHT statement (REPWEIGHTS variables;). For more information, visit https://documentation.sas.com/doc/en/pgmsascdc/v_074/statug/statug_surveyphreg_details29.htm.

Stata. Standard errors based on BRR can be generated in Stata using the 'svy' commands. The data must first be declared as complex survey data using the 'svyset' command. This command also designates variables that contain information about the survey design, and specifies the default method for variance estimation. The 'vce' option to this command should be set for "brr" ("vce(brr)") for balanced repeated replication; the weight variables are specified with the option 'brrweight(varlist).' For more information, visit https://www.stata.com.

SUDAAN. The user needs to specify the sample design as a "Balanced Repeated replication" design as well as specifying the replicate weight variables. Specifying the sample design (DESIGN = BRR) is done in the procedure call statement (i.e., PROC DESCRIPT DESIGN = BRR;). The specification of the replicate weights is done with the REPWGT statement (i.e., to produce the sampling errors for estimates use the statement: REPWGT REPW1-REPW88;). For more information, visit http://www.rti.org/sudaan.

R. Standard errors can be generated in R using function svrepdesign from the survey package. Within the arguments of the function, the user should define type = "BRR" and repweights = the replicate weight variables. For more information, visit https://r-survey.r-forge.r-project.org/survey/html/svrepdesign.html.

Approximate Sampling Errors

Although the preferred method of calculating sampling errors to reflect the aspects of the complex sample design of PSS is using Balanced Repeated Replication (BRR), it is also possible to obtain approximate standard errors by simpler methods. One such method uses design effects of some key estimates obtained from the survey. The design effect (DEFF) of a survey estimate is defined as the ratio of the variance of the estimate under the sampling design used for the survey to the variance of the estimate under simple random sampling.

In stratified sampling designs like the one used for the 2023–24 PSS, cases within a particular stratum tend to have responses that are more similar than if the cases were chosen completely at random from the population. The appropriate value of DEFF depends on the particular domain being analyzed (e.g., the DEFF for secondary/high schools is different than that for elementary/middle schools). Since each estimate has a different design effect and these may be unstable, an average DEFF was computed over different variables. Table 1 gives average values of DEFF for selected subgroups.

It is important to use estimation methods that take the sample design into account. There are a variety of variance estimation methods, and BRR is the method that was chosen in the past for

PSS. The goal of using a sample design more complex than simple random sampling is to ensure reliable estimates for subgroups of interest. Simple random sampling cannot ensure specific reliable estimates. A potential cost of using a complex sample design is increased variance compared to simple random sampling (Wolter, 1985). This cost is measured by the design effect. The approximation of a variance based on simple random sampling without fully considering the complex design may lead to incorrect conclusions, with particular risk of Type I errors: erroneously rejecting the null hypothesis and claiming statistical significance where it may not truly exist. To maintain statistical integrity, data users must use a variance estimate that accounts for the complexity of the sample design.

A simple method of obtaining the approximate standard error of an estimate from the survey is to first compute the standard error of the estimate under simple random sampling and multiply the standard error by the square root of the design effect (DEFT). An example of how to approximate the standard error of a survey mean is as follows. First, the mean should be estimated using the full sample weight and any standard statistical package like SAS or SPSS. Second, the standard error of the estimate should be obtained under simple random sampling without using weights. These simple random sample standard errors should not include any finite population correction. This unweighted standard error should then be multiplied by the square root of the average design effect to get the approximate standard error of the mean under the design.

For all schools and each category (school type, school level, enrollment size, urbanicity type, and Census regions) calculate variances for each estimate (totals for each variable of interest) under simple random sampling (SRS) and under BRR. Calculate the design effects for each estimate (totals for each variable of interest) as the ratio of BRR variances to SRS variances. Calculate the average design effect (DEFF) by averaging the multiple calculated design effects. Calculate DEFTs as the square root of each DEFF. For example, suppose that the simple random sampling standard error (unweighted) is 2.59 students in elementary/middle schools. The approximate standard error for the estimate would then be 1.71 (i.e., 2.59 students times 0.66, the DEFT for elementary/middle schools (table 1)).

Table 1. Average PSS design effects (DEFF) and square root of the average design effects (DEFT) by selected school characteristics: 2023–24

Selected school characteristic	DEFF	DEFT
Total	0.3259	0.5709
Private school type		
Catholic	0.4076	0.6384
Other religious	0.2902	0.5387
Nonsectarian	0.5679	0.7536
School level		
K-Terminal	0.3249	0.5700
Elementary/Middle	0.4350	0.6596
Secondary/High	0.3398	0.5829
Combined/Other	0.2876	0.5863
Enrollment size		
Less than 50	0.3721	0.6100
50–149	0.5179	0.7196
150–299	0.5576	0.7468
300–499	0.5569	0.7463
500–749	0.4554	0.6749
750 or more	0.5296	0.7277
Urbanicity type		
City	0.3759	0.6131
Suburban	0.6263	0.7914
Town	0.4575	0.6764
Rural	0.1405	0.3749
Region		
Northeast	0.5163	0.7185
Midwest	0.2958	0.5439
South	0.3089	0.5558
West	0.4460	0.6678

SOURCE: U.S. Department of Education, National Center for Education Statistics, Private School Universe Survey (PSS), 2023–24.

Ⅲ. Data Collection

The 2023–24 PSS data collection began in September 2023 and continued until June 2024. The 2023–24 PSS used a mailout data collection to invite schools to respond by paper questionnaire or by the internet. There were two mailing methods depending on school type, schools that do not use internet or telephones (primarily Amish and Mennonite schools) received a paper questionnaire and non-Amish schools were offered an internet response option. In addition to the mailings, non-Amish schools received emails to respond online starting September 2023 through April 2024. Follow-up of nonresponding cases was conducted first by computer-assisted telephone interviewing (CATI) and then by Census Bureau field representatives. Table 2 shows the specific data collection activities and the time frame when each occurred.

Table 2. PSS data collection schedule: 2023–24

Activity	Date(s)
Production deployment of Internet data collection	9/12/2023
1st mailing: Internet letter to non-Amish schools only	9/12/2023
Initial mailing email to schools: List frame	9/12/2023
2nd mailing: PSS questionnaires to Amish and non-Amish schools	10/3/2023
3rd mailing: Label and mail reminder postcard (Amish) and pressure sealed-mailer (non-Amish)	10/24/2023
First reminder email to schools: List frame	10/24/2023
4th mailing: PSS questionnaires to Amish and non-Amish schools	11/7/2023
Pre-CATI email to schools: List frame	12/5/2023
Begin CATI data collection for list frame	1/8/2024
Conduct no phone field follow-up	1/8/2024
Initial mailing email to schools: Area frame	1/9/2024
Pre-CATI email to schools: Area frame	1/23/2024
Begin CATI data collection for area frame	2/6/2024
Field Representative (FR) no phone field close out	2/23/2024
Regional Office (RO) no phone fields close out	3/1/2024
CATI close out: List frame and area frame	3/15/2024
Pre-field NRFU email to schools: Partial response list and area frame	3/19/2024
Pre-field NRFU email to schools: No response list and area frame	3/19/2024
Conduct field follow-up: CATI nonrespondents	4/1/2024
Final email to schools: Partial response list and area frame	4/30/2024
Final email to schools: No response list and area frame	4/30/2024
FR close out: CATI nonrespondents	5/24/2024
RO close out: CATI nonrespondents	5/31/2024
Data collection closeout	6/7/2024
5th mailing: Amish thank you postcard	6/17/2024

NOTE: Amish cases include Mennonite cases.
SOURCE: U.S. Department of Education, National Center for Education Statistics, Private School Universe Survey (PSS), 2023–24.

Data Collection Procedures

Mailings

The 2023–24 PSS data collection began on September 12, 2023, with the mailout of a letter to most private schools describing how to respond to the 2023–24 PSS online and included a User ID and password for the online option.

The second mailing of survey material was sent out on October 3, 2023, and included paper questionnaires to Amish and Mennonite schools. Additionally, non-Amish schools that were non-responsive to the invitation to complete the interview via the internet received their first paper questionnaire.

On October 24, 2023, reminder/thank you postcards were sent to all Amish and Mennonite schools while non-Amish schools that had still not responded received a reminder pressure-sealed mailer which included a User ID and password for the online option.

On November 7, 2023, all remaining non-responding Amish and Mennonite, as well as non-Amish schools, received the second and final mailing of the PSS questionnaire. The PSS mailings return rate for paper and web response was 54.2 percent prior to the start of the nonresponse follow-up efforts.[4]

Email Operation

The 2023–24 PSS email operation occurred alongside mailing attempts starting on September 12, 2023. Schools were contacted with information about the PSS and a secure link to respond online. Non-responding schools could receive up to four reminder emails with the final email sent on April 30, 2024.

Nonresponse Follow-up

In addition to the mailing and email attempts to the schools, there were also several nonresponse follow-up attempts. Census Bureau field representative conducted telephone follow-up and field follow-up for non-responding schools.

Telephone Follow-up

Telephone follow-up began on January 8, 2024, and targeted PSS schools that had not responded by mail but for which phone numbers were available as well as schools identified as area frame from the frame operation. Telephone interviewing took place at the Census Bureau's CATI facilities located in Tucson, Arizona and Jeffersonville, Indiana. CATI follow-up continued through March 15, 2024.

Field Follow-up

On January 8, 2024, the Census Bureau's six Regional Field Offices began in-person follow-up on PSS cases that did not have a telephone number listed. Follow-up of these schools continued through February 23, 2024. Following CATI closeout of list and area frame schools in March, additional follow-up of some telephone nonresponse cases was conducted by Census Bureau field representatives and continued through May 31, 2024.

Interviewer Training

CATI interviewers and Census Bureau regional office representatives conducting the PSS field follow-up received training on the content and data collection procedures for the 2023–24 PSS. CATI operator training consisted of 1.5 hours of self-study, 1 hour for paired practices, and an

[4] The return rates are unweighted return rates calculated by dividing the total number of returns (interviews, refusals, and out-of-scopes) less undeliverable as addressed/blank questionnaires by the total number of schools.

additional (optional) 1 hour to work with the survey instruments. The Tucson training had 52 staff, and the Jeffersonville training had 41 staff.

Training for field representatives consisted of 2 hours of self-study. Three hundred two field representatives completed training and took part in the two nonresponse follow-up operations.

Efforts to Increase Response Rates

Several steps were taken to maximize response rates during data collection. A prepaid business reply envelope was included in the questionnaire mailings for respondents to use when returning the completed questionnaires. The first questionnaire mailings were followed three weeks later with reminder postcards or pressure-sealed mailers. In addition, a toll-free number and an email address were provided for respondents to contact with inquiries regarding the survey. Multiple follow-up contacts were made to nonrespondents by telephone, in person, and by email.

Response Rates

The PSS evaluates response rates at the unit level and the item level. Nonresponse bias analysis is performed for results less than 85 percent.

Unit Response Rate

The 2023–24 PSS uses two measures to evaluate response, the unweighted unit response rate and the weighted unit response rate.[5] While unweighted unit response rates generally measure the proportion of cases that produce usable information for analysis, weighted unit response rates can be used to estimate the proportion of the population covered by the units that responded. For the 2023–24 PSS, the unweighted unit response rate was 76.8 percent, while the weighted response rate was 76 percent (table 3).

[5] The response rate is calculated as follows: number of responding schools divided by the number of responding schools plus the number of nonresponding schools. Out-of-scope schools are excluded from the response rate calculation.

Table 3. Number of responding and nonresponding schools, out-of-scope cases, and school response rate, by frame: 2023–24

Response status	List frame	Area frame	Total frame
Total			
Base-weighted	56,675	2,333	59,008
Unweighted	56,675	590	57,265
Response			
Base-weighted	22,324	890	23,214
Unweighted	22,324	186	22,510
Nonresponse			
Base-weighted	6,587	753	7,340
Unweighted	6,587	222	6,809
Out-of-scope[1]			
Base-weighted	27,764	691	28,455
Unweighted	27,764	182	27,946
Response rate			
Base-weighted	77.2%	54.2%	76.0%
Unweighted	77.2%	45.6%	76.8%

[1] Out-of-scope status is defined as not eligible for the PSS survey during data collection.
NOTE: The base weight is equal to the inverse of the probability of selection of the school. Detail may not sum to totals due to rounding.
SOURCE: U.S. Department of Education, National Center for Education Statistics, Private School Universe Survey (PSS), 2023–24.

Unit Nonresponse Bias Analysis

Because NCES Statistical Standard 4-4 requires analysis of unit nonresponse bias for any survey stage with a base-weighted response rate of less than 85 percent, the 2023–24 PSS data file was evaluated for potential nonresponse bias. Comparisons between the frame and respondent populations were made before and after the nonresponse weighting adjustments were applied to evaluate the extent to which the adjustments reduced or eliminated nonresponse bias. The following section explains the methodology and summarizes the conclusions.

As outlined in the NCES Statistical Standards, the degree of nonresponse bias is a function of two factors: the nonresponse rate and how much the respondents and nonrespondents differ on survey variables of interest. The mathematical formulation to estimate bias for a sample mean of variable y is given in equation 2, where $\bar{y}_T$ equals the estimated mean based on all eligible sample cases; $\bar{y}_R$ equals the estimated mean based only on respondent cases; $\bar{y}_M$ equals the estimated mean based only on nonrespondent cases; n_T equals the estimated number of cases (i.e., $n_T = n_R + n_M$); n_M equals the estimated number of nonrespondents; and n_R equals the estimated number of respondents.

Equation 2: $$B(\bar{y}_R) = \bar{y}_R - \bar{y}_T = \left(\frac{n_M}{n_T}\right)(\bar{y}_R - \bar{y}_M)$$

A variable-free estimate of the bias, referred to as a relative bias, was used to compare biases across all variables included in the analysis. The relative bias for an estimated mean using only the respondent data, $\bar{y}_R$, is calculated as given in equation 3.

Equation 3: $$RelB(\bar{y}_R) = \frac{B(\bar{y}_R)}{\bar{y}_R}$$

Relative bias was estimated for variables known for respondents and nonrespondents. There is a very limited amount of information available about nonresponding private schools. The variables used in the 2023–24 PSS unit nonresponse bias analysis were: Affiliation; Grade level of school; Enrollment; Community type; Region; and Source of the school (list or area frame).

The following steps were followed to compute the relative bias. First, the nonresponse bias was estimated and tested to determine if the bias is significant at the 5 percent level. Second, nonresponse adjustments were applied, with the variables listed above included in the nonresponse models. The nonresponse adjustments, which are included in the weights, were designed to significantly reduce, or eliminate unit nonresponse bias for variables included in the models. Third, after the weights were computed, any remaining bias was estimated for the variables listed above and statistical tests were performed to check the remaining significant nonresponse bias. For this comparison, nonresponse bias was calculated as the difference between the base-weighted sample proportion and the nonresponse-adjusted respondent proportion, which evaluates the effectiveness of each nonresponse adjustment in mitigating nonresponse bias. Schools found to be ineligible for the PSS were excluded from the analysis.

As shown in table 4, the weighting adjustments eliminated some significant bias. For all respondents, 88 percent of the variable categories were significantly biased before nonresponse weighting adjustments were made. After the nonresponse adjustments were made, 28 percent of variable categories were significantly biased. The mean estimated absolute percent relative bias was reduced by 85 percent, from 14 percent to 2 percent. Details by affiliation and reporting characteristics are presented in tables 5 and 6.

Table 4. Summary of 2023–24 PSS nonresponse bias statistics

Nonresponse bias statistic	Total (percent)
Before nonresponse adjustment[1]	
Mean estimated absolute percent relative bias	13.91
Median estimated absolute percent relative bias	8.66
Percent of variable categories significantly biased	87.50
After nonresponse adjustment[2]	
Mean estimated absolute percent relative bias	2.05
Median estimated absolute percent relative bias	0.94
Percent of variable categories significantly biased	28.13

[1] The difference between the base-weight respondents and the base-weight sample.
[2] The difference between the nonresponse adjusted respondents and the base-weight sample.
SOURCE: U.S. Department of Education, National Center for Education Statistics, Private School Universe Survey (PSS), 2023–24.

Table 5. Base-weighted PSS school frame distribution, interviewed distribution, estimated bias, percent relative bias, standard errors, and t statistic, by affiliation membership and selected characteristics: 2023–24

Selected characteristics	Weighted respondent proportions[1]	Weighted eligible sample proportions[1]	Estimated bias	Percent relative bias	Standard error of difference of proportions	T-test of significance for the difference in proportions
Affiliation membership						
List frame						
Catholic, Parochial	6.36	5.28	1.08	16.97	0.02	58.34
Catholic, Diocesan	11.53	9.73	1.79	15.56	0.03	53.72
Catholic, Private	3.81	4.68	−0.86	−22.68	0.01	−63.97
Amish	4.02	4.76	−0.73	−18.20	0.01	−53.74
Assembly of God	0.72	0.59	0.13	17.61	0.00	60.40
Baptist	4.95	4.44	0.51	10.28	0.01	35.77
Episcopal	1.21	1.12	0.08	6.93	0.00	24.06
Jewish	3.49	3.91	−0.42	−12.05	0.01	−37.70
Lutheran Church, Missouri Synod	3.26	2.69	0.57	17.40	0.01	59.71
Wisconsin Evangelical Lutheran Synod	1.30	1.03	0.27	20.75	0.00	70.18
Mennonite	1.15	1.54	−0.39	−33.74	0.00	−84.34
Pentecostal	0.84	1.57	−0.73	−86.07	0.01	−127.21
Seventh-Day Adventist	2.44	2.17	0.27	11.00	0.01	38.27
Other religious	19.61	19.22	0.39	1.99	0.06	6.82
Nonsectarian, regular	13.04	15.45	−2.41	−18.49	0.04	−54.44
Nonsectarian, special emphasis	12.34	10.93	1.41	11.45	0.04	39.81
Nonsectarian, special education	6.10	5.51	0.58	9.58	0.02	33.33
Area frame only						
Catholic	0.58	0.78	−0.19	−33.10	0.12	−1.66
Other religious	2.25	2.96	−0.71	−31.59	0.25	−2.85
Nonsectarian	1.00	1.64	−0.64	−63.91	0.12	−5.25
School level						
K-terminal	10.67	11.18	−0.51	−4.79	0.10	−5.01
Elementary/Middle	48.60	46.89	1.71	3.51	0.17	10.24
Secondary/High	11.76	11.45	0.31	2.66	0.06	5.51
Combined/Other	28.96	30.47	−1.51	−5.21	0.16	−9.64
Enrollment - list frame only[2]						
Less than 50	36.94	38.17	−1.23	−3.33	†	†
50–149	27.76	26.88	0.89	3.20	†	†
150–299	18.65	18.05	0.60	3.20	†	†
300–499	9.79	9.49	0.30	3.06	†	†
500–749	3.83	4.03	−0.20	−5.30	†	†
750 or more	3.03	3.38	−0.35	−11.55	†	†
Region						
Northeast	20.62	21.68	−1.06	−5.14	0.07	−14.72
Midwest	23.91	22.99	0.92	3.85	0.14	6.35
South	35.69	35.56	0.13	0.38	0.16	0.86
West	19.78	19.78	0.00	0.02	0.09	0.05

Selected characteristics	Weighted respondent proportions[1]	Weighted eligible sample proportions[1]	Estimated bias	Percent relative bias	Standard error of difference of proportions	T-test of significance for the difference in proportions
Community type						
City	35.69	35.56	0.13	0.37	0.14	0.96
Suburban	35.67	35.41	0.26	0.72	0.12	2.18
Town	8.78	8.10	0.68	7.73	0.14	4.91
Rural	19.86	20.93	−1.07	−5.38	0.15	−7.24

† Not applicable as list frame schools were not selected through the sampling process and do not receive replicate weights.

[1] Weighted using the base weights.

[2] The enrollment size bias is calculated for list frame only, because the enrollment size is not assigned for area frame non-respondents.

NOTE: Detail may not sum to totals because of rounding or missing values in cells with too few sample cases.

SOURCE: U.S. Department of Education, National Center for Education Statistics, Private School Universe Survey (PSS), 2023–24.

Table 6. Final weighted PSS school frame distribution, interviewed distribution, estimated bias, percent relative bias, standard errors, and t statistic, by affiliation membership and selected characteristics: 2023–24

Selected characteristics	Weighted respondent proportions[1]	Weighted eligible sample proportions[1]	Estimated bias	Percent relative bias	Standard error of difference of proportions	T-test of significance for the difference in proportions
Affiliation membership						
List frame						
Catholic, Parochial	5.28	5.28	−0.01	−0.10	0.02	−0.22
Catholic, Diocesan	9.73	9.73	0.00	−0.04	0.04	−0.10
Catholic, Private	4.56	4.68	−0.12	−2.65	0.02	−6.16
Amish	4.75	4.76	−0.01	−0.17	0.02	−0.40
Assembly of God	0.60	0.59	0.01	1.57	0.00	3.56
Baptist	4.56	4.44	0.11	2.47	0.02	5.57
Episcopal	1.17	1.12	0.05	4.25	0.01	9.48
Jewish	3.88	3.91	−0.03	−0.86	0.02	−1.97
Lutheran Church, Missouri Synod	2.71	2.69	0.02	0.65	0.01	1.47
Wisconsin Evangelical Lutheran Synod	1.03	1.03	0.00	0.30	0.00	0.68
Mennonite	1.54	1.54	0.00	−0.15	0.01	−0.34
Pentecostal	1.56	1.57	−0.02	−1.02	0.01	−2.36
Seventh-Day Adventist	2.17	2.17	0.00	−0.11	0.01	−0.26
Other religious	19.22	19.22	0.00	0.00	0.08	0.00
Nonsectarian, regular	15.45	15.45	0.00	0.00	0.07	0.00
Nonsectarian, special emphasis	10.93	10.93	0.00	0.00	0.05	0.00
Nonsectarian, special education	5.51	5.51	0.00	0.00	0.02	0.00
Area frame only						
Catholic	0.82	0.78	0.04	5.09	0.16	0.27
Other religious	3.15	2.96	0.19	6.17	0.39	0.49
Nonsectarian	1.40	1.64	−0.24	−16.88	0.13	−1.86
School level						
K-terminal	10.98	11.18	−0.20	−1.83	0.13	−1.61

Selected characteristics	Weighted respondent proportions[1]	Weighted eligible sample proportions[1]	Estimated bias	Percent relative bias	Standard error of difference of proportions	T-test of significance for the difference in proportions
Elementary/Middle	46.62	46.89	−0.28	−0.60	0.24	−1.17
Secondary/High	12.03	11.45	0.58	4.84	0.08	7.45
Combined/Other	30.37	30.47	−0.10	−0.34	0.29	−0.36
Enrollment - list frame only[2]						
Less than 50	38.65	38.17	0.48	1.25	†	†
50–149	27.11	26.88	0.23	0.85	†	†
150–299	17.84	18.05	−0.21	−1.20	†	†
300–499	9.37	9.49	−0.12	−1.24	†	†
500–749	3.87	4.03	−0.17	−4.32	†	†
750 or more	3.17	3.38	−0.22	−6.83	†	†
Region						
Northeast	21.03	21.68	−0.65	−3.10	0.10	−6.43
Midwest	23.01	22.99	0.02	0.10	0.15	0.15
South	36.38	35.56	0.83	2.28	0.25	3.29
West	19.58	19.78	−0.20	−1.02	0.14	−1.47
Community type						
City	35.34	35.56	−0.22	−0.62	0.20	−1.11
Suburban	35.25	35.41	−0.17	−0.47	0.16	−1.00
Town	8.46	8.10	0.36	4.29	0.32	1.13
Rural	20.95	20.93	0.02	0.10	0.14	0.14

† Not applicable as list frame schools were not selected through the sampling process and do not receive replicate weights.

[1] Weighted using the nonresponse adjusted (final) weights.

[2] Weighted using the base weights.

[3] The enrollment size bias is calculated for list frame only, because the enrollment size is not assigned for area frame non-respondents.

NOTE: Detail may not sum to totals because of rounding or missing values in cells with too few sample cases.

SOURCE: U.S. Department of Education, National Center for Education Statistics, Private School Universe Survey (PSS), 2023–24.

Item Response Rate

Just as schools sometimes chose not to respond to the 2023–24 PSS survey request, those that did respond occasionally chose not to answer all of the survey questions. Unweighted item response rates are calculated by dividing the number of schools responding to an item by the number of schools asked to reply to an item. Weighted item-level response rates in the 2023–24 PSS were generally high, ranging from 89.9 to 100 percent (table 7). Because the weighted item response rates were all greater than 85 percent, no item-level bias analyses were performed.

Table 7. PSS item response rates for private schools: 2023–24

Item number	Unweighted response rate	Final weighted response rate	Item number	Unweighted response rate	Final weighted response rate
Q4A_GRD	99.9	99.9	Q6A	96.9	96.4
Q4A_ENRL	99.8	99.8	Q6B	96.8	96.3
Q4B_GRD	99.9	99.9	Q6C	96.8	96.4
Q4B_ENRL	99.8	99.8	Q6D	96.8	96.3
Q4C_GRD	99.9	99.9	Q6E	96.7	96.2
Q4C_ENRL	99.7	99.7	Q6F	96.7	96.3
Q4D_GRD	99.9	99.9	Q6G	96.7	96.3
Q4D_ENRL	99.9	99.9	Q7A	99.8	99.7
Q4E_GRD	99.9	99.9	Q7B	92.9	93.0
Q4E_ENRL	100.0	100.0	Q8A	100.0	100.0
Q4F_GRD	99.9	99.9	Q8B	99.8	99.8
Q4F_ENRL	99.7	99.7	Q8C	98.6	98.4
Q4G_GRD	99.9	99.9	Q8D	94.1	94.0
Q4G_ENRL	99.7	99.7	Q9A	99.5	99.5
Q4H_GRD	99.9	99.9	Q9B	99.0	98.9
Q4H_ENRL	99.7	99.7	Q10_FT	99.5	99.4
Q4I_GRD	99.9	99.9	Q10_>3/4	99.4	99.3
Q4I_ENRL	99.7	99.7	Q10_>1/2	99.4	99.3
Q4J_GRD	99.9	99.9	Q10_>1/4	99.4	99.3
Q4J_ENRL	99.7	99.7	Q10_<1/4	99.4	99.3
Q4K_GRD	99.9	99.9	Q10_TOTAL	99.6	99.6
Q4K_ENRL	99.7	99.7	Q11A	97.2	97.2
Q4L_GRD	99.9	99.9	Q11B	90.4	90.5
Q4L_ENRL	99.7	99.6	Q11C	89.9	89.9
Q4M_GRD	99.9	99.9	Q12	99.4	99.3
Q4M_ENRL	99.7	99.6	Q13A	99.6	99.6
Q4N_GRD	99.9	99.9	Q13B	100.0	100.0
Q4N_ENRL	99.7	99.7	Q14A	100.0	100.0
Q4O_GRD	99.9	99.9	Q14B	96.9	96.9
Q4O_ENRL	99.7	99.6	Q14C	99.6	99.6
Q4P_GRD	99.9	99.9	Q14C_CATHOLIC	99.3	99.1
Q4P_ENRL	99.7	99.7	Q15	99.9	99.9
Q4Q_GRD	99.9	99.9	Q16	98.0	97.8
Q4Q_ENRL	99.7	99.6	Q17	99.2	99.1
Q5_TOTAL	99.5	99.5	Q18	99.5	99.4

NOTE: The final weight is equal to the product of the base weight and a nonresponse adjustment factor for the school.
SOURCE: U.S. Department of Education, National Center for Education Statistics, Private School Universe Survey (PSS), 2023–24.

Ⅳ. Data Preparation

Data from the interviews are captured into one file and went through several edits as preparation for release including computer pre-edit, computer edit, and imputation.

Questionnaire Check-in and Data Capture

Paper questionnaires returned by individual respondents and those completed by field representatives during field follow-up were sent to the Census Bureau's National Processing Center (NPC) in Jeffersonville, Indiana. Upon receipt, codes were assigned to each questionnaire to indicate its status (e.g., complete interview, refusal, or out-of-scope). Then the questionnaires were grouped into batches for data capture.

The 2023–24 PSS data were captured using a combination of manual data keying and imaging technology, using the Integrated Computer Assisted Data Entry (iCADE) system. The questionnaires were disassembled, and each page was scanned. Clerical staff reviewed and resolved data issues due to system errors such as the inability to read a barcode, inability to recognize a page corner point, or checkbox ambiguities (e.g., marks outside checkbox, scratch outs, random marks on a page). Each imaged response was presented to a keying operator and all fields except the checkboxes were keyed from image (KFI). The checkboxes were captured using Optical Mark Recognition (OMR). All non-blank data fields were 100 percent KFI verified.

After the PSS questionnaire data were captured, the output files from the different sources (keyed, Internet, and CATI) were merged into one file. Corrections to the name, address, and telephone numbers supplied by the respondents were then applied to the PSS file. These corrections were keyed, either by NPC or during the CATI interviews. All addresses were then checked with the National Change of Address (NCOA) software, a commercial software package that corrects and standardizes addresses.

Computer Pre-Edit

After data capture, the data were processed through a pre-edit designed to identify respondent errors in variables needed for determining interview status or imputation. This pre-edit program generated lists of cases where there were potential data problems. Census Bureau staff reviewed the computer record of cases that met each pre-edit rejection condition. Whenever possible, unacceptable entries were corrected by the analyst using information reported in other questionnaire items or information from other sources. Unacceptable entries that could not be corrected were blanked in the data record. These procedures are referred to as "pre-edits" because staff, rather than computer algorithms, handled computer-identified problems.

Computer Edit

After pre-edit corrections were made, the file was submitted to a completely automated computer edit. This edit consisted of a consistency edit, a blanking edit, and a logic edit. The consistency edit identified inconsistent entries within each record. The fields compared could be within items (e.g., if the response in item 8a, whether any students were enrolled in 12th grade last year, was "No," but the second part of the question reported 12th graders enrolled the previous year) or between items (e.g., if kindergarten was reported in item 4c and the response to item 9a indicated

that the school did not offer kindergarten). Entries were corrected for those inconsistencies that could be resolved; for those that could not be resolved, the entries were deleted. The blanking edit assigned the "not answered" code to items that should have been answered but were not and deleted entries to items that should not have been answered.

In the logic edit, data from the 2021–22 PSS were used to fill the items with missing values whenever possible. For example, if the type of school was not reported in item 12 and it had been reported on the 2021–22 PSS, the 2021–22 response was copied to the 2023–24 record.

In addition to filling items where values were missing, some inconsistencies between students by race and total enrollment items were corrected by ratio adjustment during the logic edit. For records where the number of students in item 6 (students by race) did not equal the enrollment in item 5 (total enrollment), excluding prekindergarten, the item 6 entries were adjusted to be consistent with item 5; that is, each entry in item 6 was multiplied by the ratio of the student count in item 6 to the enrollment in item 5, excluding prekindergarten. For a record where items 11a-11c (class type) were all answered as no, but students were enrolled in the school, the school was set to offer in-person classes (11a).

Imputation

After the edit processing was complete, there were missing values within some records classified as interviews. These were cases where the respondent had not answered some applicable questionnaire items (and data for these items were not added in the pre-edit, consistency edit, or logic edit) or the response had been deleted during editing. Values were imputed to the missing data during imputation, which is described below.

Donor Imputation

Items missing after edits were imputed using donor imputation. Values were created by extracting data from the record for a sample case (donor) with similar characteristics, using a procedure known as the "sequential nearest neighbor hot deck" (Kalton and Kasprzyk 1982, 1986; Kalton 1983; Little and Rubin 1987; Madow, Olkin, and Rubin 1983). In order to match incomplete records to those with complete data, "imputation" variables that identify certain characteristics of the school that were deemed to be important to the reporting of the data in each item (e.g., religious affiliation, enrollment, school level of instruction) were used. Items were grouped according to the perceived relevance of the imputation variables to the data collected by the item. For example, school level of instruction was used for matching incomplete records and donors to fill item 16 (length of school year) but was not used for item 6 (students by race).

Items 13b (located in a private home that is used primarily as a family residence), 14a and 14c (religious orientation or purpose, and religious orientation or affiliation), and 15 (association membership) did not go through the donor imputation program. If, after the logic edit, item 13b or 15 was still incomplete, the assumption was made for item 13b that the school was not located in a private home and for item 15 that the school did not belong to any associations. If values were still missing for items 14a and 14c after the logic edit, the records were reviewed and imputed by an analyst.

Analyst Imputation

There were cases where the imputation failed to create a value because (1) there was no suitable donor, or (2) the value imputed was invalid because it was outside the acceptable range for the item or was inconsistent with other data on the same record, or (3) the religious orientation or purpose, or the religious orientation or affiliation, was not reported (items 14a and 14c) and no previous PSS information was available. Staff reviewed the data record, sample file record, and the questionnaire to impute a value consistent with the information from those sources.

Imputation Flags

The imputation flags indicate the imputation method used, such as donor imputation or analyst imputation. The codes used for imputation flags are described in section V.

Ⅴ. Guide to the Public-Use Data File and Codebook

The 2023–24 PSS public-use data file contains data from all 22,510 completed interviews. The public-use data are available in SAS, SPSS, and in Microsoft Excel Comma Separated Values (CSV) file format. The public-use materials also include syntax files, codebook, record layout, survey questionnaire, frame data file, and frame file data dictionary.

The 2023–24 PSS public-use data does not contain one variable (Item 8c – P355) and its associated imputation flag variable (F_P355), which are on the PSS restricted-use data file. Variable P355 is the number of students graduating 12th grade in the prior school year.

Categories of Variables

Variables on the PSS data files fall into one of five categories: frame, survey, weighting, imputation flag, and composite variables. The purpose of these categories is to help the user better understand what types of variables are included on the files and what the sources were for the variables.

Variables were classified as frame variables if they were drawn from the 2021–22 PSS file or based on the 2023–24 PSS frame update. Examples of frame variables include the school's permanent identification number (PPIN) or the ZIP code of the school's mailing address (PZIP).

Survey variables are the actual variables drawn from the questionnaire responses. Each item on the questionnaire has a small number printed to the left. This number is the source code. A single letter "P" was added to the beginning of the source code to create the corresponding variable name. For example, item 5a has a source code of 305. On the data file, the variable name for this item is P305.

There are two types of weighting variables on the file. The first is the final weight for the school, and the second includes the 88 replicate weights. The final weight adjusts for nonresponse and the sampling rates of the area-frame PSUs and is used so that estimates represent the population of private schools. The replicate weights are used as a set to generate standard errors for estimates. On the file, the final weight is named PFNLWT, and the replicate weights are REPW1 to REPW88.

The imputation flags identify whether a survey item was imputed for missing data. All survey variables have a corresponding imputation flag that indicates whether a value was imputed and, if so, what method was used. All survey imputation flags begin with "F_" and are followed by the name of the variable. For example, the imputation flag for P135 is F_P135. The flag values represent the type of imputation method used and are as follows: 0 = Value not imputed; 4 = Imputed by using a donor value; 5 = Data adjusted by analyst during review.

Composite variables are based on survey variables, frame variables, or other created variables. These variables are frequently used in NCES publications and have been added to the files to facilitate data analysis. A list of the composite variables included in the file is presented in the next section with an explanation of how they were derived.

Composite Variables

CSOURCE – Method of interview:

Purpose: To provide a summary classification of survey response method.

General explanation: Based on survey response through one of four means.

Values:

1 = Mail

2 = Internet

3 = Regional Office follow-up

4 = CATI (computer-assisted telephone interview)

DIOCESE – Diocese or archdiocese of Catholic schools:

Purpose: To provide the Diocese in which a Catholic school is located.

General explanation: Based on the Catholic school's physical location (PCNTY).

Values:

See codebook.

FRAME – universe frame to which school belongs:

Purpose: To provide an indicator of which universe frame was used to identify a school for participation in the survey summary classification of how the school responded to the survey.

General explanation: Based on which universe frame was used to include a school for participation

Values:

LIST = school is part of list frame

AREA = school is part of area frame

GRADE2 – Level of instruction (Four Categories):

Purpose: To provide a recode of LEVEL4 with a separate K-terminal category for schools that offers kindergarten, transitional kindergarten, or transitional first grade

General explanation: Based on LEVEL4 and grades offered (P135, P145, P155, P165, P175, P185, P195, P205, P215, P225, P235, P245, P255, P265, P275, P285, and P295).

Values:

0 = K-terminal. (LEVEL4 = 1 and (P155 = 1 or P165 = 1 or P175 = 1) and (P185-P295 ≠ 1))

1 = Elementary/Middle. (LEVEL4 = 1)

2 = Secondary/High. (LEVEL4 = 2)

3 = Combined/Other. (LEVEL4 = 3)

HIGR2024 – highest grade level in school:

Purpose: To provide a measure of the highest grade level in the school.

General explanation: The highest reported grade level in the school based on the grades offered (P135, P145, P155, P165, P175, P185, P195, P205, P215, P225, P235, P245, P255, P265, P275, P285, and P295).

Values:

1 = All Ungraded (P135 = 1 and P145-P295 = 2)

2 = Highest grade in school is prekindergarten (P145 = 1 and P155-P295 = 2)

3 = Highest grade in school is kindergarten (P155 = 1 and P165-P295 = 2)

4 = Highest grade in school is transitional kindergarten (P165 = 1 and P175-P295 = 2)

5 = Highest grade in school is transitional 1st grade (P175 = 1 and P185-P295 = 2)

6 = Highest grade in school is 1st grade (P185 = 1 and P195-P295 = 2)

7 = Highest grade in school is 2nd grade (P195 = 1 and P205-P295 = 2)

8 = Highest grade in school is 3rd grade (P205 = 1 and P215-P295 = 2)

9 = Highest grade in school is 4th grade (P215 = 1 and P225-P295 = 2)

10 = Highest grade in school is 5th grade (P225 = 1 and P235-P295 = 2)

11 = Highest grade in school is 6th grade (P235 = 1 and P245-P295 = 2)

12 = Highest grade in school is 7th grade (P245 = 1 and P255-P295 = 2)

13 = Highest grade in school is 8th grade (P255 = 1 and P265-P295 = 2)

14 = Highest grade in school is 9th grade (P265 = 1 and P275-P295 = 2)

15 = Highest grade in school is 10th grade (P275 = 1 and P285-P295 = 2)

16 = Highest grade in school is 11th grade (P285 = 1 and P295 = 2)

17 = Highest grade in school is 12th grade (P295 = 1)

LEVEL – grade level of school (used prior to 2019–20 PSS):

Purpose: To provide a summary measure of the grades provided by the school.

General explanation: Based on the grades offered (P135, P145, P155, P165, P175, P185, P195, P205, P215, P225, P235, P245, P255, P265, P275, P285, and P295).

Values:

1 = Elementary: A school is elementary if it has one or more of grades K–6 and does not have any grade higher than grade 8.

2 = Secondary*:* A school is secondary if it has one or more of grades 7–12 and does not have any grade lower than grade 7.

3 = Combined*:* A school is classified as combined if it has one or more of grades K–6 and one or more of grades 9–12. Schools in which all students are ungraded (i.e., not classified by standard grade levels) are also classified as combined.

LEVEL2 – grade level of school (used with 2019–20 PSS):

Purpose: To provide a summary measure of the grades provided by the school.

General explanation: Based on the grades offered (P135, P145, P155, P165, P175, P185, P195, P205, P215, P225, P235, P245, P255, P265, P275, P285, and P295).

Values:

1 = Elementary/Middle. A school is elementary/middle if it has students in more of grades PK through 4 than in higher grades or that enroll students in more of grades 5 through 8 than in higher or lower grades.

2 = Secondary/High. A school is secondary/high if it enrolls students in more of grades 9 through 12 than in lower grades; or if a school offers the same number of grades 9 through 12 as the school offers 5 through 8 without offering any K through 4 grades.

3 = Combined/Other. A school is classified as combined/other if it has one or more of grades K–6 and one or more of grades 9–12; for example, schools with grades K–12, 6–12, 6–9, or 1–12 are classified as having combined grades. Schools in which all students are ungraded (i.e., not classified by standard grade levels) are also classified as combined.

LEVEL3 – grade level of school:

Purpose: Special re-calculation of LEVEL4 to recode elementary schools that had 12th grade students in 2022–23 to combined/other.

General explanation: Based on LEVEL4 and P345

Values:

1 = Elementary/Middle. (LEVEL4 = 1 and P345 = 2)

2 = Secondary/High. (LEVEL4 = 2)

3 = Combined/Other. (LEVEL4 = 3 or (LEVEL4 = 1 and P345 = 1))

LEVEL4 – grade level of school (used beginning with 2021–22 PSS):

Purpose: To provide a summary measure of the grades provided by the school.

General explanation: Based on the grades offered (P135, P145, P155, P165, P175, P185, P195, P205, P215, P225, P235, P245, P255, P265, P275, P285, and P295).

Values:

1 = Elementary/Middle. A school offers more grades in K-4 compared to grades in 5-8; or a school offers more grades in 5-8 compared to grades in K-4 and grades in 9-11.

2 = Secondary/High. A school offers more or the same grades in 9-11 compared to grades in 5-8; or a school offers grade 12 and offers more or the same grades in 9-11 compared to grades in 5-8; or a school offers grades 9-11 and K, but does not offer grades 1-8 or grade 12; or a school offers grade 12 alone; or a school offers grade 12 and K, but does not offer grades 1-8.

3 = A school offers ungraded only; or a school offers at least one grade K-4 and at least one grade 5-8 and at least one grade 9-12.

LATITUDE24 – latitude:

Purpose: To provide a measure of a school's location.

General explanation: Based on the school's physical location.

Values:

The 3 digits to the left of the decimal represent the number of degrees from the equator; the 6 digits to the right of the decimal represent the fraction of the next degree carried out to six decimal places.

LOGR2024 – lowest grade level in school:

Purpose: To provide a measure of the lowest grade level in the school.

General explanation: Based on the grades offered (P135, P145, P155, P165, P175, P185, P195, P205, P215, P225, P235, P245, P255, P265, P275, P285, and P295).

Values:

1 = All Ungraded (P135 =1 and P145-P295 = 2)

2 = Lowest grade in school is prekindergarten (P145 = 1 and P135 = 2)

3 = Lowest grade in school is kindergarten (P155 = 1 and P135-P145 = 2)

4 = Lowest grade in school is transitional kindergarten (P165 = 1 and P135-P155 =2)

5 = Lowest grade in school is transitional 1st grade (P175 = 1 and P135-P165 = 2)

6 = Lowest grade in school is 1st grade (P185 = 1 and P135-P175 = 2)

7 = Lowest grade in school is 2nd grade (P195 = 1 and P135-P185 = 2)

8 = Lowest grade in school is 3rd grade (P205 = 1 and P135-P195 = 2)

9 = Lowest grade in school is 4th grade (P215 = 1 and P135-P205 = 2)

10 = Lowest grade in school is 5th grade (P225 = 1 and P135-P215 = 2)

11 = Lowest grade in school is 6th grade (P235 = 1 and P135-P225 = 2)

12 = Lowest grade in school is 7th grade (P245 = 1 and P135-P235 = 2)

13 = Lowest grade in school is 8th grade (P255 = 1 and P135-P245 = 2)

14 = Lowest grade in school is 9th grade (P265 = 1 and P135-P255 = 2)

15 = Lowest grade in school is 10th grade (P275 = 1 and P135-P265 = 2)

16 = Lowest grade in school is 11th grade (P285 = 1 and P135-P275 = 2)

17 = Lowest grade in school is 12th grade (P295 = 1 and P135-P285 = 2)

LONGITUDE24 – longitude:

Purpose: To provide a measure of a school's location.

General explanation: Based on the school's physical location.

Values:

The 3 digits to the left of the decimal represent the number of degrees from the prime meridian; the 6 digits to the right of the decimal represent the fraction of the next degree carried out to six decimal places.

MALES – number of K-12 and ungraded male students in the school:

Purpose: To provide a measure for the total number of male students in the school.

General explanation: Based on the coeducational status of the school (P335), the number of male students in coeducational schools (P340), and NUMSTUDS.

Values:

MALES = P340 (if P340 ≥ 0), or

0 (if P335 = 2), or

NUMSTUDS (if P335 = 3)

NUMSTUDS – number of K-12 and ungraded students in the school:

Purpose: To provide a summary measure of the non-prekindergarten enrollment in the school.

General explanation: Based on the grades-by-grade enrollment (P140, P160, P170, P180, P190, P200, P210, P220, P230, P240, P250, P260, P270, P280, P290, and P300).

Values:

NUMSTUDS = P140 + P160 + P170 + P180 + P190 + P200 + P210 + P220 + P230 + P240 + P250 + P260 + P270 + P280 + P290 + P300

NUMTEACH – number of Full-Time-Equivalent (FTE) teachers in the school:

Purpose: To provide a measure of the number of FTE teachers in the school.

General explanation: Based on the number of teachers in each full-time and part-time work status category (P385, P390, P395, P400, and P405)

Values:

NUMTEACH = P385 + (.875 x P390) + (.625 x P395) + (.375 x P400) + (.125 x P405)

ORIENT – school's religious or nonsectarian orientation:

Purpose: To provide a detailed religious and nonsectarian classification of private schools.

General explanation: Based on the school's religious orientation (P430) or affiliation (P440):

Values:

1 = Roman Catholic (P440 = 1)

2 = African Methodist Episcopal (P440 = 2)

3 = Amish (P440 = 3)

4 = Assembly of God (P440 = 4)

5 = Baptist (P440 = 5)

6 = Brethren (P440 =6)

7 = Calvinist (P440 = 7)

8 = Christian, no specific denomination (P440 = 8)

9 = Church of Christ (P440 = 9)

10 = Church of God (P440 = 10)

11 = Church of God in Christ (P440 = 11)

12 = Church of the Nazarene (P440 = 12)

13 = Disciples of Christ (P440 = 13)

14 = Episcopal (P440 = 14)

15 = Friends (P440 = 15)

16 = Greek Orthodox (P440 = 16)

17 = Islamic (P440 = 17)

18 = Jewish (P440 = 18)

19 = Latter Day Saints (P440 = 19)

20 = Lutheran Church - Missouri Synod (P440 = 20)

21 = Evangelical Lutheran Church in America (P440 = 21)

22 = Wisconsin Evangelical Lutheran Synod (P440 = 22)

23 = Other Lutheran (P440 = 23)

24 = Mennonite (P440 = 24)

25 = Methodist (P440 = 25)

26 = Pentecostal (P440 = 26)

27 = Presbyterian (P440 = 27)

28 = Seventh-day Adventist (P440 = 28)

29 = Other (P440 = 29)

30 = Nonsectarian (P430 = 2)

P_ASIAN – percentage of K-12 and ungraded students who are Asian, not of Hispanic or Latino origin:

Purpose: To provide a measure of the percentage of the school's students who are Asian.

General explanation: Based on the number of Asian students (P316) and total K-12 and ungraded enrollment (NUMSTUDS).

Values:

P_ASIAN = P316 / NUMSTUDS

P_BLACK – Percentage of K-12 and ungraded students who are Black or African American, not of Hispanic or Latino origin:

Purpose: To provide a measure of the percentage of the school's students who are Black or African American.

General explanation: Based on the number of Black or African American students (P325) and total K-12 and ungraded enrollment (NUMSTUDS).

Values:

P_BLACK = P325 / NUMSTUDS

P_HISP – percentage of K-12 and ungraded students who are Hispanic or Latino, regardless of race:

Purpose: To provide a measure of the percentage of the school's students who are Hispanic or Latino.

General explanation: Based on the number of Hispanic or Latino students (P320) and total K-12 and ungraded enrollment (NUMSTUDS).

Values:

P_HISP = P320 / NUMSTUDS

P_INDIAN – percentage of K-12 and ungraded students who are American Indian or Alaska Native, not of Hispanic or Latino origin:

Purpose: To provide a measure of the percentage of the school's students who are American Indian or Alaska Native.

General explanation: Based on the number of American Indian or Alaska Native students (P310) and total K-12 and ungraded enrollment (NUMSTUDS).

Values:

P_INDIAN = P310 / NUMSTUDS

P_PACIFIC – percentage of K-12 and ungraded students who are Native Hawaiian or other Pacific Islander, not of Hispanic or Latino origin:

Purpose: To provide a measure of the percentage of the school's students who are Native Hawaiian or other Pacific Islander.

General explanation: Based on the number of Native Hawaiian or other Pacific Islander students (P318) and total K-12 and ungraded enrollment (NUMSTUDS).

Values:

P_PACIFIC = P318 / NUMSTUDS

P_TR – percentage of K-12 and ungraded students who are two or more races, not of Hispanic or Latino origin:

Purpose: To provide a measure of the percentage of the school's students who are two or more races.

General explanation: Based on the number of students who are two or more races (P332) and total K-12 and ungraded enrollment (NUMSTUDS).

Values:

P_TR = P332 / NUMSTUDS

P_WHITE – percentage of K-12 and ungraded students who are White, not of Hispanic or Latino origin:

Purpose: To provide a measure of the percentage of the school's students who are White.

General explanation: Based on the number of White students (P330) and total K-12 and ungraded enrollment (NUMSTUDS).

Values:

P_WHITE = P330 / NUMSTUDS

PCNTY24 – ANSI county code and state code:

Purpose: To provide a composite indicator of a school's county and state code

General explanation: Based on the physical location of the school (ANSI county code) and state FIPS.

Values:

ANSI county code (PCNTY) + State code (PSTANSI)

REGION – geographic region of school:

Purpose: To provide a summary measure of the school's physical location.

General explanation: Based on the state of the school's physical location (PSTANSI).

Values:

1 = Northeast: Maine, New Hampshire, Vermont, Massachusetts, Rhode Island, Connecticut, New York, New Jersey, and Pennsylvania;

2 = Midwest: Ohio, Indiana, Illinois, Michigan, Wisconsin, Minnesota, Iowa, Missouri, North Dakota, South Dakota, Nebraska, and Kansas;

3 = South: Delaware, Maryland, District of Columbia, Virginia, West Virginia, North Carolina, South Carolina, Georgia, Florida, Kentucky, Tennessee, Alabama, Mississippi, Arkansas, Louisiana, Oklahoma, and Texas; and

4 = West: Montana, Idaho, Wyoming, Colorado, New Mexico, Arizona, Utah, Nevada, Washington, Oregon, California, Alaska, and Hawaii.

RELIG – school's religious affiliation:

Purpose: To provide a summary classification for private schools.

General explanation: Based on the school's religious orientation and affiliation (P430 and P440)

Values:

1 = Catholic (TYPOLOGY in (1, 2, 3))

2 = Other religious (TYPOLOGY in (4, 5, 6))

3 = Nonsectarian (TYPOLOGY in (7, 8, 9))

SIZE – categorized number of K-12 and ungraded students in the school:

Purpose: To provide a summary classification for private school enrollment.

General explanation: Based on the school's K-12 and ungraded enrollment (NUMSTUDS)

Values:

1 = Less than 50 students (NUMSTUDS = 1 – 49)

2 = 50–149 students (NUMSTUDS = 50 – 149)

3 = 150–299 students (NUMSTUDS = 150 – 299)

4 = 300–499 students (NUMSTUDS = 300 – 499)

5 = 500–749 students (NUMSTUDS = 500 – 749)

6 = 750 students or more (NUMSTUDS greater than 749)

S_KG – total number of kindergarten students in the school:

Purpose: To provide a total count of number of kindergarten students in the school.

General explanation: Based on the school's kindergarten (P160), transitional kindergarten (P170) and transitional first grade enrollment (P180).

Values:

S_KG = P160 + P170 + P180

STTCH_RT – student/teacher ratio:

Purpose: To provide a measure of the student/teacher ratio for K-12 and ungraded levels.

General explanation: Based on the school's FTE teachers (NUMTEACH) and total K-12 and ungraded enrollment (NUMSTUDS).

Values:

STTCH_RT = NUMSTUDS / NUMTEACH

TOTHRS – length of school day in hours:

Purpose: To provide a measure for the total length of the school day.

General explanation: Based on the length of the school day in hours (P650) and minutes (P655)

Values:

TOTHRS = P650 + P655/60

TYPOLOGY – NCES 9-category typology:

Purpose: To provide a summary classification for private schools.

General explanation: Based on the school's religious affiliation (RELIG and P445), association membership (P450-P640), and program emphasis (P415)

Values:

1 = Catholic, parochial – Catholic schools with a parochial or inter-parochial governance structure (RELIG =1 and P445 = 1)

2 = Catholic, diocesan – Catholic schools with a Diocesan governance structure (RELIG = 1 and P445=2)

3 = Catholic, private – Catholic schools with a private governance structure (RELIG = 1 and P445 = 3)

4 = Other Religious, Conservative Christian – Other religious schools with membership in at least one of four associations: Accelerated Christian Education, American Association of Christian Schools, Association of Christian Schools International, or Oral Roberts University Education Fellowship (RELIG = 2 and P455, P460, P465, or P520 = 1)

5 = Other Religious, affiliated – Other religious schools not classified as Conservative Christian with membership in at least one of sixteen associations—Agudath Israel of America, Association of Christian Teachers and Schools, Association of Classical Christian Schools, Christian Schools International, Council of Islamic Schools in North America, Evangelical Lutheran Education Association, Friends Council on Education, General Conference of the Seventh-Day Adventist Church, Islamic School League of America, Jesuit Schools Network, National Alliance of Christian Schools, National Association of Episcopal Schools, National Christian School Association, National Society for Hebrew Day Schools, Prizmah: Center for Jewish Day Schools —or indicating membership in "other religious school associations." (RELIG = 2, TYPOLOGY ne 4, and P458, P467, P468, P470, P475, P480, P485, P490, P492, P495, P500, P510, P515, P523, P530 or P535 = 1)

6 = Other Religious, unaffiliated – Other religious schools not classified as Conservative Christian or affiliated (RELIG = 2 and TYPOLOGY ne 4 or 5)

7 = Nonsectarian, Regular – Nonsectarian schools with a regular or early childhood program emphasis (RELIG = 3 and P415 = 1 or 7)

8 = Nonsectarian, special emphasis – Nonsectarian schools with a Montessori, career/technical/vocational, alternative, or special program emphasis (RELIG = 3 and P415 = 2, 3, 5, or 6)

9 = Nonsectarian, special education – Nonsectarian schools with a special education program emphasis (RELIG = 3 and P415 =4)

UCOMMTYP – urban-centric community type:

Purpose: To provide a measure of a school's community type relative to populous areas :

General explanation: Based on the urban-centric local code (ULOCALE).

Values:

1 = City (ULOCALE24 = 11, 12, or 13)

2 = Suburb (ULOCALE24 = 21, 22, or 23)

3 = Town (ULOCALE24 = 31, 32, or 33)

4 = Rural (ULOCALE24 = 41, 42, or 43)

ULOCALE24 – urban-centric locale code:

Purpose: To provide a measure of a school's location relative to populous areas.

General explanation: Based on the school's physical address or the mailing address, if the physical address is not reported

(https://nces.ed.gov/programs/edge/docs/EDGE_GEOCODE_PRIVATESCH_FILEDOC.pdf)

Values:

11 = City, Large: Territory inside an urbanized area and inside a principal city with population of 250,000 or more.

12 = City, Midsize: Territory inside an urbanized area and inside a principal city with population less than 250,000 and greater than or equal to 100,000.

13 = City, Small: Territory inside an urbanized area and inside a principal city with population less than 100,000.

21 = Suburb, Large: Territory outside a principal city and inside an urbanized area with population of 250,000 or more.

22 = Suburb, Midsize: Territory outside a principal city and inside an urbanized area with population less than 250,000 and greater than or equal to 100,000.

23 = Suburb, Small: Territory outside a principal city and inside an urbanized area with population less than 100,000.

31 = Town, Fringe: Territory inside an urban cluster that is less than or equal to 10 miles from an urbanized area.

32 = Town, Distant: Territory inside an urban cluster that is more than 10 miles and less than or equal to 35 miles from an urbanized area.

33 = Town, Remote: Territory inside an urban cluster that is more than 35 miles of an urbanized area.

41 = Rural, Fringe: Census-defined rural territory that is less than or equal to 5 miles from an urbanized area, as well as rural territory that is less than or equal to 2.5 miles from an urban cluster.

42 = Rural, Distant: Census-defined rural territory that is more than 5 miles but less than or equal to 25 miles from an urbanized area, as well as rural territory that is more than 2.5 miles but less than or equal to 10 miles from an urban cluster.

43 = Rural, Remote: Census-defined rural territory that is more than 25 miles from an urbanized area and is also more than 10 miles from an urban cluster.

The variable names, descriptions, and descriptive statistics for all variables on the data file are contained in the codebook.

Unique School Identifier

Each school record on the 2023–24 PSS file contains a unique school identifier, the permanent identification number (PPIN). PPIN is a unique 8-character code assigned when a private school is first included in PSS. PPIN can be used to match a school's 2023–24 PSS record to those of other years.

VI. User Notes and Cautions

This section provides some caveats and considerations for analysts when using the 2023–24 PSS data.

Locale Code (Urbanicity)

The 2023–24 PSS data files do not contain the 8-category locale code (LOCALE) that was included on the 2005–06 and earlier PSS files. The 2023–24 files do contain the 12-category locale code (ULOCALE24), which has been included on the PSS data files since the 2003–04 survey. The final two characters of the ULOCALE variable in PSS files correspond to the data collection year (e.g., 2024).

Race/Ethnicity

Beginning with the 2009–10 PSS, race/ethnicity data (Item 6) were collected following the Department of Education's October 2007 guidance on collecting and reporting race and ethnicity data. The seven race/ethnicity categories used were Hispanic or Latino, regardless of race (P320); and for those not of Hispanic or Latino origin, American Indian or Alaska Native (P310), Asian (P316), Black or African American (P325), Native Hawaiian or other Pacific Islander (P318), White (P330), and Two or more races (P332). In earlier PSS collections, race/ethnicity data were collected using five categories; Asians and Pacific Islanders were combined in one category - Asian or Pacific Islander, and there was no category for those of two or more races.

Standard Error

For PSS, only the area frame contributes to the standard error. Since all list-frame cases are included in the PSS, the list frame component of the standard error is always zero (0).

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